Aditya Bhoite
22108A0033

Page No.
Date

REM Assignment - 4

Q.1]

Principle of Hydropower generation : Hydropower works by using flowing water to spin the turbine. Water stored at height (dam) has potential energy.

Types of Hydropower Plants -

Storage (Dam - based) : stores water and releases when needed.

Run of River - Uses Natural Riverflow with large storage.

Pumped Storage - Pumps water up and reuses it (energy storage system).

Q2]

→ Biomass refers to organic materials that can be used as fuel. Common sources include agricultural ~~wastte~~ waste like crop residues, animal waste such as dung, forest residues like wood, even municipal organic waste. These materials are easily available and help in waste management while producing energy.

To convert biomass into usable energy several technologies are used. Combustion involves directly burning biomass to generate heat or electricity. Anaerobic digestion produces gas from organic waste which can be used as fuel. Gasification converts ~~of~~ biomass into combustible gas, while pyrolysis produces bio-oil and charcoal in the absence of oxygen. Additionally, biofuels like ethanol and biodiesel are produced from crops.

Page No.
Date

Q3]
→ 1) Energy Management is the process of using energy efficiently to reduce waste, lower costs, and improve overall performance.
2) It involves monitoring energy usage, identifying inefficiencies, and implementing improvements.
3) for Industries, effective energy management leads to cost Savings and reduced environmental Impact.
4) To Implement an Energy Management System (EMS), the first step is conducting an energy audit to understand current consumption patterns.
5) Based on this clear goals and targets for Energy Savings are set.

Q4]
→ 1) Renewable Energy projects require a high initial investments, mainly due to the equipment and installation cost. 2) However once installed, the operating costs are relatively low since ~~they~~ they do not rely on fuel
3) Overtime, this results in significant costs savings.
4) Government incentives such as subsidies and tax benefit further improve the financial feasiblity of these projects.
5) In the long run, renewable energy proves to be economically Sustainable, as it provides Stable energy costs and reduces dependence on fluctuating fossil fuel prices.

Page No.
Date

Q.5]

→ 1.) Renewable energy plays a crucial role in sustainable development by reducing pollution and conserving natural resources.

2.) It ensures that future generations have access to clean and reliable energy.

3.) By lowering carbon emissions, it helps combat climate change

4.) In Organizations, renewable energy supports better decisions-making.

5.) Managers can use this information to plan investments, improve operations, and meet sustainability targets.